

Lemma 10.1 Let $Q \in \mathbb{R}^{n \times m}$ where $m \leq n$ such that its columns are orthonormal. Then $Q^T Q = I$

Proof: As $(Q^T Q)_{i,j} = \langle Q^T_{i,\cdot}, Q_{\cdot,j} \rangle$
 $= \langle Q_{\cdot,i}, Q_{\cdot,j} \rangle$
 $= \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$

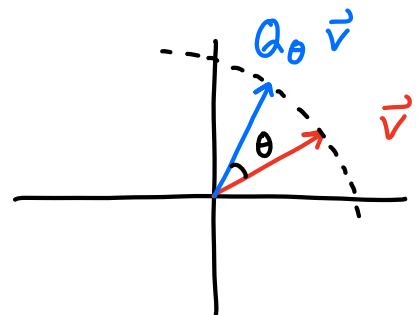
we have that $Q^T Q = I$ as desired.

Corollary 10.2 If $Q \in \mathbb{R}^{m \times m}$ has orthonormal columns, then $Q Q^T = I$. Hence Q is invertible with inverse of $Q^{-1} = Q^T$. These matrices are said to be orthogonal.

Examples of orthogonal matrices on $\mathbb{R}^2$

i. Counter - Clockwise rotation by θ

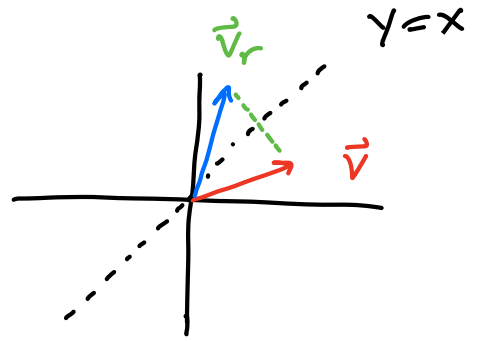
$$Q_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$



ii. Reflection across $\vec{v}$

The matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

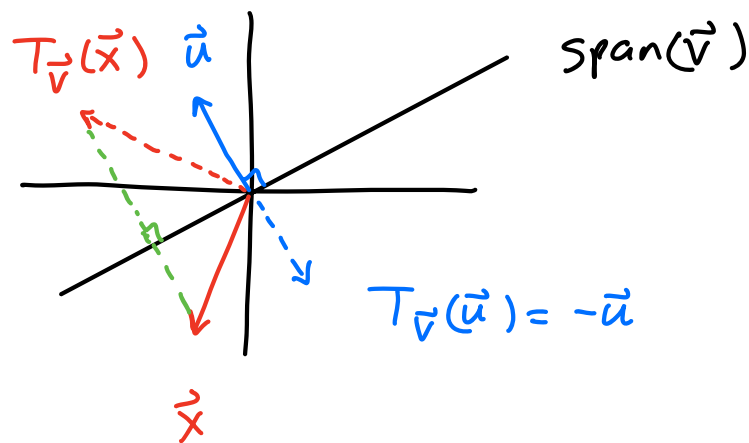
represents the reflection
across $y=x$



Exercise:

- 1) Find the matrix representation of the reflection across $\vec{v} \in \mathbb{R}^2$ ($\vec{v} \neq \vec{0}$)

Picture:



Let $\vec{u} \in \mathbb{R}^2$ be such that $\langle \vec{u}, \vec{v} \rangle = 0$

and set $T_{\vec{v}}(\vec{v}) = \vec{v}$ and $T_{\vec{v}}(\vec{u}) = -\vec{u}$.

Now $[T_{\vec{v}}]_{\{\vec{e}_i\}} = [\vec{v} \quad -\vec{u}] [\vec{v} \quad \vec{u}]^{-1}$

and letting $\vec{v} = \begin{bmatrix} a \\ b \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} -b \\ a \end{bmatrix}$

so that $\langle \vec{v}, \vec{u} \rangle = 0$ we have that

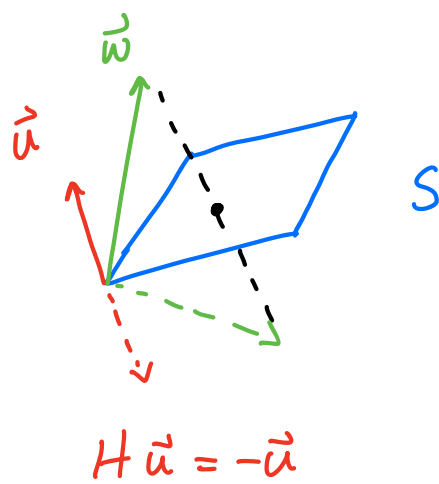
$$\begin{aligned} [{}^T \vec{v}]_{\{\vec{e}_i\}} &= \begin{bmatrix} a & b \\ b & -a \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix}^{-1} \\ &= \frac{1}{a^2+b^2} \begin{bmatrix} a & b \\ b & -a \end{bmatrix} \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \\ &= \frac{1}{a^2+b^2} \begin{bmatrix} a^2-b^2 & 2ab \\ 2ab & b^2-a^2 \end{bmatrix} \\ &= \frac{1}{a^2+b^2} \begin{bmatrix} a^2+b^2-2b^2 & 2ab \\ 2ab & a^2+b^2-2a^2 \end{bmatrix} \\ &= I - \frac{1}{a^2+b^2} \begin{bmatrix} 2b^2 & -2ab \\ -2ab & 2a^2 \end{bmatrix} \\ &= I - 2 \left(\frac{1}{\sqrt{a^2+b^2}} \begin{bmatrix} -b \\ a \end{bmatrix} \right) \left(\frac{1}{\sqrt{a^2+b^2}} \begin{bmatrix} -b & a \end{bmatrix} \right) \\ &= I - 2 \frac{\vec{u}}{\|\vec{u}\|} \frac{\vec{u}^T}{\|\vec{u}\|} \end{aligned}$$

This is called a Householder matrix

Theorem 10.3 The reflection across the hyperplane

$S = \{ \vec{v} \in \mathbb{R}^n : \langle \vec{v}, \vec{u} \rangle = 0 \}$ where $\|\vec{u}\| = 1$ is

given by $H = I - 2\vec{u}\vec{u}^T$



Theorem 10.4 Every orthogonal 2×2 matrix is a product of a rotation and a reflection.

Proof: Let Q be orthogonal so that

$$Q = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \quad \text{or} \quad Q = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$$

$\det 1$ $\det -1$

where $a^2 + b^2 = 1$. In case 1, Q is a rotation by $\theta = \arctan \frac{b}{a}$. In case 2,

$$Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} b & -a \\ a & b \end{bmatrix} \quad \text{which represents}$$

a reflection across the diagonal and a rotation by $\theta = \arctan \frac{a}{b}$.

10.5 Properties of Orthogonal matrices

i. If Q_1 and Q_2 are orthogonal, then $Q_1 Q_2$ is orthogonal

ii. Orthogonal matrices are isometries

$$\|Q \vec{x}\| = \|\vec{x}\|$$

iii. Orthogonal matrices are conformal (angle preserving)

$$\frac{\langle Q \vec{x}, Q \vec{y} \rangle}{\|Q \vec{x}\| \|Q \vec{y}\|} = \frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\| \|\vec{y}\|}$$

Theorem 10.6 $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ is conformal (angle preserving) iff $[T] = \lambda Q$ where $\lambda \in \mathbb{R}$ and Q is orthogonal.

$$M_{i,j} = \langle \vec{e}_i, M \vec{e}_j \rangle$$

Proof: Assume that T is conformal and let $(\Rightarrow)$ $A = [T]_{\{\vec{e}_i\}}$. Since $\{\vec{e}_i\}$ is orthonormal we have that $\langle T \vec{e}_i, T \vec{e}_j \rangle = 0$ for $i \neq j$.
Now as $0 = \langle A \vec{e}_i, A \vec{e}_j \rangle$
$$= \langle \vec{e}_i, A^T A \vec{e}_j \rangle$$

for $i \neq j$ we have that $A^T A$ is diagonal.

Now let $1 \leq i, j \leq n$ be chosen arb.
 Since $\langle \vec{e}_i + \vec{e}_j, \vec{e}_i - \vec{e}_j \rangle = \|\vec{e}_i\|^2 - \|\vec{e}_j\|^2 = 0$,
 (i ≠ j)

$\langle T(\vec{e}_i + \vec{e}_j), T(\vec{e}_i - \vec{e}_j) \rangle = 0$ and thus

$$\begin{aligned} 0 &= (A\vec{e}_i + A\vec{e}_j)^T (A\vec{e}_i - A\vec{e}_j) \\ &= \vec{e}_i^T (A^T A) \vec{e}_i - \vec{e}_j^T (A^T A) \vec{e}_j \end{aligned}$$

Hence $A^T A = \lambda^2 I$ and thus $A = \lambda Q$
 for some orthogonal matrix.

The set of orthogonal matrices is known as
 the general orthogonal group

$$O(n, \mathbb{R}) = \{ Q \in \mathbb{R}^{n \times n} : Q^T Q = I \}$$

and represents conformal maps on $\mathbb{R}^n$ that
 fix exactly one point. The special orthogonal
 group $SO(n, \mathbb{R})$ consists of orthogonal matrices
 of determinant 1 (reflection free)

Linear Functionals (Axler 6.B)

Def 10.7 Let V be an inner product space over F . A linear functional is a linear map $\varphi \in \mathcal{L}(V, F)$

Examples:

i. $\varphi: \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $\varphi(\vec{x}) = \langle \vec{x}, \vec{v} \rangle$ for some fixed $\vec{v} \in \mathbb{R}^n$

ii. $\varphi: \mathcal{P}_n(\mathbb{R}) \rightarrow \mathbb{R}$ defined by

$$\varphi(f) = \int_{-1}^1 \underbrace{f(t) \cos k\pi t}_{\text{FCS coefficient}} dt$$

While not obvious, there exists $g \in \mathcal{P}_n(\mathbb{R})$ such that $\varphi(f) = \langle f, g \rangle$ where

$$\langle f, g \rangle = \int_{-1}^1 f \cdot g$$

Theorem 10.8 (Riesz Representation Theorem)

Suppose that V is a finite dimensional, inner product space over F and φ is a linear functional on V . Then there is a unique

$\vec{u} \in V$ s.t. $\varphi(\vec{v}) = \langle \vec{v}, \vec{u} \rangle$ for every $\vec{v} \in V$.

Proof: let $\{\vec{e}_1, \dots, \vec{e}_n\}$ be an orthonormal basis for V . Now for any $\vec{v} \in V$ we have that $\vec{v} = \sum_{i=1}^n \langle \vec{v}, \vec{e}_i \rangle \vec{e}_i$ and thus

$$\varphi(\vec{v}) = \sum_{i=1}^n \langle \vec{v}, \vec{e}_i \rangle \varphi(\vec{e}_i) \quad \text{where } \varphi(\vec{e}_i) \in F$$

$$\begin{aligned} \lambda \langle \vec{x}, \vec{y} \rangle &= \langle \lambda \vec{x}, \vec{y} \rangle &= \sum_{i=1}^n \langle \lambda \vec{x}, \overline{\varphi(\vec{e}_i)} \vec{e}_i \rangle \\ &= \langle \vec{x}, \overline{\lambda} \vec{y} \rangle &= \langle \vec{x}, \sum_{i=1}^n \overline{\varphi(\vec{e}_i)} \vec{e}_i \rangle \end{aligned}$$

The proof of uniqueness follows a similar pattern as earlier proofs:

letting $\vec{u}_1, \vec{u}_2$ be such that

$$\varphi(\vec{v}) = \langle \vec{v}, \vec{u}_1 \rangle = \langle \vec{v}, \vec{u}_2 \rangle$$

for all $\vec{v}$, we have that for all $\vec{v}$

$$\langle \vec{v}, \vec{u}_1 - \vec{u}_2 \rangle = 0$$

and thus $\vec{u}_1 - \vec{u}_2 = \vec{0}$.

If $\{\vec{e}_1, \dots, \vec{e}_n\}$ is an orthonormal basis for V and φ is a linear functional, then

$$\varphi(\vec{v}) = \langle \vec{v}, \vec{u} \rangle \text{ where } \vec{u} = \sum_{i=1}^n \overline{\varphi(\vec{e}_i)} \vec{e}_i$$

Exercise:

3) Find $g \in P_2(\mathbb{R})$ such that

$$\int_{-1}^1 \overset{\varphi(f)}{f(t) \cos \pi t} = \int_{-1}^1 \overset{\langle f, g \rangle}{f(t) g(t)} dt$$

From before we found orthogonal basis

$f_0 = 1$, $f_1 = x$, $f_2 = x^2 - \frac{1}{3}$. Now unitizing

these we obtain orthonormal basis:

$$e_0 = \frac{f_0}{\sqrt{\langle f_0, f_0 \rangle}} = \frac{1}{\sqrt{\int_{-1}^1 1}} = \frac{1}{\sqrt{2}}$$

$$e_1 = \frac{f_1}{\sqrt{\langle f_1, f_1 \rangle}} = \frac{x}{\sqrt{\int_{-1}^1 x^2}} = \frac{x}{\sqrt{2/3}} = \sqrt{\frac{3}{2}} x$$

$$e_2 = \frac{f_2}{\sqrt{\langle f_2, f_2 \rangle}} = \frac{x^2 - \frac{1}{3}}{\sqrt{\int_{-1}^1 (x^2 - \frac{1}{3})^2}} = \sqrt{\frac{45}{8}} (x^2 - \frac{1}{3})$$

Thus $g = \varphi(e_0)e_0 + \varphi(e_1)e_1 + \varphi(e_2)e_2$

$$= \left(\frac{1}{\sqrt{2}} \int_{-1}^1 \cancel{\cos \pi t}^0 \right) \frac{1}{\sqrt{2}} +$$

$$\left(\sqrt{\frac{3}{2}} \int_{-1}^1 \cancel{t \cos \pi t}^0_{\text{odd}} \right) \sqrt{\frac{3}{2}} t +$$

$$\left(\sqrt{\frac{45}{8}} \int_{-1}^1 (t^2 - \frac{1}{3}) \cos \pi t \right) \sqrt{\frac{45}{8}} (t^2 - \frac{1}{3})$$

$$= \frac{45}{8} (t^2 - \frac{1}{3}) \left(\frac{-4}{\pi^2} \right)$$

$$= \frac{-45}{2\pi^2} (t^2 - \frac{1}{3})$$

u	dv
$t^2 - \frac{1}{3}$	$\cos \pi t$